

Industrial Internship Report

on

“Quality Prediction in a Mining Process”

Predicting % Silica Concentrate in an Iron Ore Flotation Plant Using Machine Learning

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Program

B.Tech. Computer Science & Engineering – AI/ML

Internship Duration

6 Weeks

Industrial Partner

UniConverge Technologies Pvt Ltd (UCT)

Facilitated by

upskill Campus / The IoT Academy

Executive Summary

This report presents the work carried out during a six-week industrial internship on the problem “Quality Prediction in a Mining Process”. The project focuses on building a machine-learning-based soft sensor for predicting the percentage of silica (% Silica Concentrate) in the final iron ore concentrate of a froth flotation plant.

The assigned problem is based on real industrial process data. The public dataset contains approximately 737,453 observations and 24 variables covering March to September 2017. The data combines feed-quality measurements, reagent flows, pulp conditions, flotation-column air flows and levels, and final laboratory quality measurements. The dataset description states that some variables are sampled every 20 seconds while other measurements are available hourly; the final silica measurement is a laboratory target and is therefore delayed relative to many process signals.

The internship work was structured into six progressive stages: industrial/domain understanding; data preparation and exploratory analysis; time alignment and feature engineering; baseline regression modelling; model tuning and forecasting-horizon experiments; and final validation, documentation and presentation. The proposed workflow compares interpretable regression baselines with tree-based ensemble methods and provides a path toward sequence models such as LSTM for future work.

A key academic and industrial outcome is the recognition that this is a time-series soft-sensor problem rather than a simple static regression problem. Published research on related flotation data demonstrates the usefulness of lagged process variables and sequence/deep-learning approaches for estimating silica and iron concentrate quality. The report therefore emphasizes leakage-safe time splits, delayed-target alignment, MAE/RMSE/R² evaluation, feature importance, and future real-time deployment.

Item	Project Detail
Domain	Mining / Mineral Processing / Industrial AI
Process	Iron ore froth flotation
Prediction task	Regression / soft-sensor prediction
Target	% Silica Concentrate
Dataset size	~737,453 rows × 24 columns
Time period	March–September 2017
Main sampling	20-second sensor measurements; hourly laboratory quality
Core tools	Python, Pandas, NumPy, Matplotlib, Scikit-learn, XGBoost; optional TensorFlow/Keras

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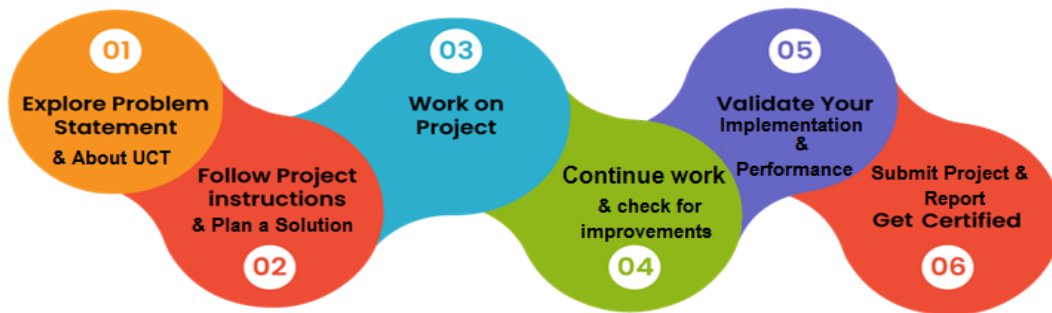
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1 Preface

The six-week industrial internship was undertaken to bridge academic machine-learning knowledge with a practical industrial quality-prediction problem. The project provided exposure to a real process domain in which measurements arrive at different sampling rates and the desired quality variable is obtained from laboratory analysis with delay.

The assigned project, Quality Prediction in a Mining Process, addresses prediction of silica impurity in iron ore concentrate. The objective is not merely to fit a regression equation, but to develop a reliable analytical workflow that can transform process measurements into an early indication of product quality. Such a soft sensor can support operators and engineers by providing an estimate before the laboratory result becomes available.

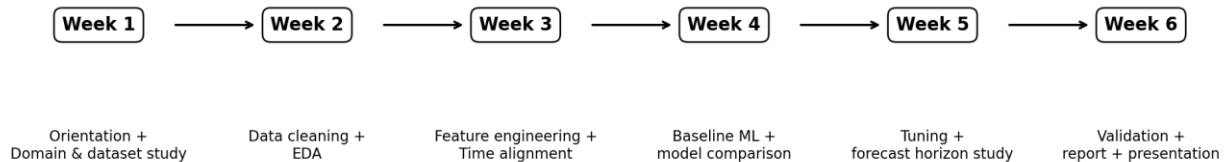
The internship was organized as a six-week progression from problem understanding to final documentation. Each week had a specific technical outcome, while the overall work followed the lifecycle of an industrial data-science project: understand → prepare → explore → engineer → model → validate → communicate.



The internship also strengthened practical skills in Python programming, data cleaning, time-series analysis, regression modelling, model evaluation, technical documentation, research reading and presentation. The experience demonstrated the importance of data quality, temporal alignment and domain knowledge in industrial AI.

I express my gratitude to the internship coordinators, mentors and academic faculty who supported the learning process. I also acknowledge the dataset contributors and researchers whose published work made it possible to study the flotation quality-prediction problem in an academically grounded manner.

1.1 Six-Week Internship Schedule



Week 1 – Orientation, Problem Understanding and Research

- Understood the internship objectives, industrial context and expected academic deliverables.
- Studied froth flotation at a conceptual level: feed, reagents, air flow, column levels, concentrate and tailings.
- Read the supplied problem statement and identified the three core questions: minute-level prediction, forecast horizon, and prediction without % Iron Concentrate.
- Reviewed related research on soft sensors, Random Forest/Gradient Boosting, LSTM and flotation-quality prediction.
- Defined the target as % Silica Concentrate and documented potential data leakage risks.

Week 2 – Dataset Acquisition, Cleaning and Exploratory Data Analysis

- Studied the 24-column industrial dataset and the difference between 20-second process signals and delayed laboratory measurements.
- Converted the date/time field to a proper datetime representation and standardized decimal-comma numeric fields.
- Checked missing values, duplicate observations, inconsistent types, outliers and basic descriptive statistics.
- Explored distributions, time trends and correlations among feed quality, reagent flow, pulp conditions, air flow/levels and concentrate quality.
- Prepared a clean modelling table and documented the data dictionary.

Week 3 – Time Alignment and Feature Engineering

- Investigated temporal alignment between process measurements and the laboratory target.
- Created resampling concepts for 1-minute and 1-hour modelling windows.
- Designed lag features and rolling statistics so that only information available before the prediction timestamp is used.
- Prepared feature groups: feed quality, reagent variables, pulp variables, flotation air flow, flotation levels and optional % Iron Concentrate.
- Defined two modelling tracks: with % Iron Concentrate and without it, to test dependence on the highly related quality variable.

Week 4 – Baseline Machine Learning and Model Comparison

- Established simple baselines such as mean prediction and linear regression.
- Planned and implemented tree-based regression models including Decision Tree, Random Forest and Gradient Boosting/XGBoost.
- Used train/validation/test separation with chronological ordering where forecasting is evaluated.
- Compared MAE, RMSE and R^2 and inspected residuals and feature importance.
- Documented the trade-off between accuracy, interpretability and computational cost.

Week 5 – Hyperparameter Tuning and Forecast-Horizon Experiments

- Performed the design of hyperparameter searches for the strongest baseline models.
- Investigated how prediction performance changes when the target is forecast ahead by different time horizons.
- Compared models using the % Iron Concentrate variable with models that exclude it.
- Considered sequence models such as LSTM for temporal dependencies and reviewed published findings to guide future implementation.
- Prepared an evaluation matrix suitable for industrial decision-making rather than relying on a single accuracy number.

Week 6 – Validation, Documentation, Deployment Concept and Presentation

- Consolidated preprocessing, feature engineering, model evaluation and research findings.
- Prepared the final system architecture and a soft-sensor deployment concept.
- Documented constraints, limitations, data leakage risks and future improvements.
- Prepared the internship report, presentation and project explanation for academic evaluation.
- Reflected on technical learning, industrial relevance, teamwork/communication and future career applications of AI/ML.

2 Introduction

Industrial process plants generate large volumes of sensor data, but the final product-quality variable may still depend on laboratory measurements. This creates a time gap between the actual process state and the availability of a confirmed quality value. Machine learning can be used to construct a soft sensor: an analytical estimator that uses readily available process measurements to approximate a delayed quality measurement.

Froth flotation is a mineral-processing method used to separate valuable minerals from unwanted material based on differences in surface properties. In the assigned mining scenario, the objective is to improve iron concentrate quality by reducing unwanted silica in the final concentrate. The supplied dataset captures the process at multiple time scales, making it particularly suitable for studying time-series regression and predictive quality monitoring.

The Kaggle dataset description reports that the first column is the time/date, the following columns contain feed-quality and process information, and the last column is % Silica Concentrate. The dataset page also frames the industrial questions around predicting silica more frequently, forecasting ahead, and reducing dependence on the correlated % Iron Concentrate variable.

2.1 About UniConverge Technologies Pvt Ltd

UniConverge Technologies Pvt Ltd (UCT) is described in the supplied internship template as a company established in 2013 and working in digital transformation and industrial solutions, with a focus on sustainability and return on investment.

The template identifies technologies used by UCT for industrial solutions, including Internet of Things, cyber security, cloud computing, machine learning, communication technologies such as 4G/5G/LoraWAN, Java full stack, Python and front-end technologies.

The supplied template also describes UCT Insight as an IoT platform supporting protocols such as MQTT, CoAP, HTTP, Modbus TCP and OPC UA, and describes Factory Watch as a smart-factory platform for production/asset monitoring, OEE, predictive maintenance and digital-twin-oriented scaling. These capabilities are relevant to the internship because a machine-learning soft sensor can ultimately be connected to an industrial IoT monitoring layer.

The internship project therefore fits the broader industrial-AI workflow: collect process data → clean and synchronize it → generate predictive features → estimate quality → expose the prediction to operators or downstream systems.

2.2 About upskill Campus and The IoT Academy

According to the supplied internship template, upskill Campus, The IoT Academy and UCT collaborated to facilitate the internship process. The IoT Academy is described as the EdTech division of UCT and as a platform running executive certification programs in collaboration with academic institutions.

The internship structure combined guided learning with an industry-oriented problem statement. This was important because the project required both technical machine-learning work and the ability to explain an engineering problem, its constraints, its proposed solution and its potential industrial value.

2.3 Objectives

- Understand the industrial flotation process and the role of silica as an impurity in iron concentrate.
- Prepare and analyze the real-world time-series dataset containing process and quality variables.
- Develop a machine-learning regression workflow for predicting % Silica Concentrate.
- Investigate whether minute-level prediction is feasible after suitable time alignment and resampling.
- Study how far ahead silica concentration can be forecast without unacceptable loss of predictive quality.
- Evaluate the effect of excluding % Iron Concentrate from the feature set.
- Compare regression models using MAE, RMSE and R^2 and examine feature importance.
- Develop an academic-quality system design and a future real-time deployment concept.

2.4 Research Background and References

Research on flotation soft sensors shows that the problem is strongly time-dependent. A 2018 industrial soft-sensor study described the difficulty caused by laboratory latency and compared Random Forest, Gradient Boosted Trees and Multi-Layer Perceptron approaches. A 2020 study on a manufacturing froth-flotation process used a stacked LSTM architecture to model hourly process variables and reported improved modelling capability compared with a Random Forest baseline. Another study, FlotationNet, explored a customized deep-learning architecture for iron and silica purity prediction and highlighted the difficulty of modelling silica accurately.

These studies support the project design choice to treat the task as a time-series soft-sensor problem, use lagged inputs, evaluate forecasting horizons and avoid random data leakage across time.

2.5 Glossary

Term	Meaning
% Silica Concentrate	Percentage of silica impurity in the final iron ore concentrate; project target.
% Iron Concentrate	Percentage of iron in the final concentrate; a strongly related quality variable and an optional feature.
Froth Flotation	Mineral separation process using differences in surface properties and air bubbles.
Soft Sensor	A model that estimates a hard-to-measure or delayed process variable using other measurable signals.
Lag Feature	A past value of a variable used to predict a future/current target.
RMSE	Root Mean Squared Error; penalizes larger errors more strongly.
MAE	Mean Absolute Error; average absolute prediction error.
R^2	Coefficient of determination; proportion of target variance explained by the model.
Data Leakage	Use of information during training that would not be available at prediction time.

3 Problem Statement

The assigned problem is to predict the percentage of silica in the final iron ore concentrate using measurements available from the flotation process. The target quality value is obtained from laboratory analysis, creating a delay between process operation and confirmed product quality.

The central engineering question is whether machine-learning models can act as a soft sensor and provide an earlier estimate of silica concentration. Earlier information could allow engineers to investigate process deviations and adjust operating conditions before a delayed laboratory result confirms the problem.

The problem is challenging because the data is multivariate and time-dependent, variables have different sampling frequencies, process dynamics introduce delays, and some quality variables are highly correlated. A model can appear highly accurate if it accidentally uses future information; therefore, temporal validation is a critical requirement.

3.1 Project Questions

- Is it possible to predict % Silica Concentrate at approximately minute-level resolution?
- How many steps/hours ahead can % Silica Concentrate be predicted with acceptable error?
- Can % Silica Concentrate be predicted without using % Iron Concentrate?
- Which process variables contribute most to silica prediction?
- Which regression model offers the best balance of predictive quality, stability, interpretability and computation?

3.1 Dataset Description and Range

The project uses the “Quality Prediction in a Mining Process” / Mining Process Flotation Plant Database. The public dataset is described as real industrial data from a flotation plant. The dataset contains approximately 737,453 records and 24 columns, with measurements spanning March to September 2017. The Kaggle competition version separates March–August as training data and September as test data.

Dataset Property	Value / Description
Rows	Approximately 737,453
Columns	24
Data type	Multivariate, time-series industrial data
Time span	March 2017 – September 2017
Process sampling	Some variables sampled every 20 seconds
Quality sampling	Final laboratory quality reported hourly
Target	% Silica Concentrate
Related quality variable	% Iron Concentrate
Training/test framing	Kaggle competition: March–August training, September test

Selected published descriptive ranges are shown below. They are included to document the scale of the input variables and should be re-generated from the exact CSV version used for any final experimental run.

Variable	Min	Max	Unit / Interpretation
% Iron Feed	42.7	65.8	% Fe
% Silica Feed	1.3	33.4	% SiO ₂
Starch Flow	0	6300.2	m ³ /h
Amina Flow	241.7	739.5	m ³ /h
Ore Pulp Flow	376.2	418.6	t/h
Ore Pulp pH	8.8	10.8	pH
Ore Pulp Density	1.5	1.9	t/m ³
FC 01 Air Flow	175.5	373.9	Nm ³ /h
FC 02 Air Flow	175.2	376.0	Nm ³ /h

The published descriptive statistics also illustrate why preprocessing matters: numerical values in the source CSV use comma decimal notation in many copies of the dataset, while the date field must be parsed as a datetime variable. A robust pipeline should therefore specify decimal parsing explicitly, validate numeric conversion and record any rows removed or imputed.

4 Existing and Proposed Solution

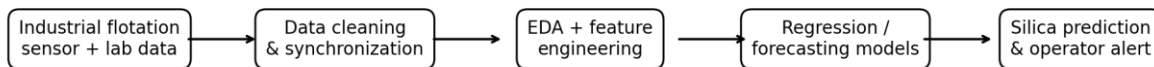
4.1 Existing Approaches

The conventional approach relies on laboratory analysis of the final concentrate. This provides an authoritative quality measurement but introduces operational latency. Research literature demonstrates the use of soft sensors, classical machine learning and deep learning to estimate silica or concentrate purity from process variables.

Approach	Strength	Limitation
Laboratory analysis	Direct quality measurement; authoritative	Delayed; cannot continuously provide a prediction
Linear regression	Simple, interpretable, fast	May not capture nonlinear flotation relationships
Decision Tree	Interpretable nonlinear rules	Can overfit without constraints
Random Forest	Strong nonlinear baseline; robust	Larger model; less transparent than linear models
Gradient Boosting / XGBoost	High predictive power on tabular data	Requires tuning; can overfit/leak if validation is wrong
LSTM / sequence model	Can learn temporal dependencies	More complex; requires careful sequence construction and validation

4.2 Proposed Solution

The proposed solution is a leakage-safe, time-aware machine-learning pipeline that converts industrial process measurements into a continuous estimate of % Silica Concentrate. The workflow begins with data validation and time synchronization, followed by exploratory analysis, feature engineering, model training and chronological evaluation.



Target: % Silica Concentrate | Time-series regression / soft-sensor concept

Two feature configurations are proposed. Configuration A includes % Iron Concentrate as a feature to establish an upper-performance reference. Configuration B removes % Iron Concentrate and uses only process/feed variables, providing a more operationally useful soft sensor because it avoids dependence on a delayed final-quality measurement.

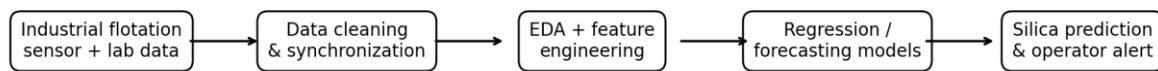
4.3 Value Addition

- Transforms delayed laboratory quality information into an earlier predictive signal.
- Introduces explicit temporal alignment and lag features rather than treating the dataset as ordinary tabular data.
- Tests minute-level and multi-hour forecasting questions separately.
- Measures the effect of removing % Iron Concentrate to reduce target leakage and operational dependence.
- Produces interpretable model diagnostics such as feature importance and residual analysis.
- Provides a deployment-ready architecture concept that can later connect to an industrial IoT dashboard.

5 Proposed Design / Model

The proposed architecture separates data engineering, modelling and monitoring layers. This separation makes it possible to replace the regression algorithm without redesigning the entire pipeline.

5.1 High Level Diagram



Target: % Silica Concentrate | Time-series regression / soft-sensor concept

Figure 1: High-level system flow for the proposed mining quality prediction soft sensor.

5.2 Low Level Data Flow

1. Raw CSV / plant historian export is loaded into the preprocessing layer.
2. Timestamp and numeric formatting are standardized.
3. Process variables are resampled and aligned to the target prediction frequency.
4. Lag and rolling features are generated using only past observations.
5. The feature matrix is passed through a preprocessing/model pipeline.
6. The trained regressor predicts % Silica Concentrate.
7. Prediction error and model health statistics are monitored.
8. The prediction can be exposed through a dashboard or industrial IoT application.

5.3 Model and Evaluation Interfaces

Interface	Input	Output
Preprocessing interface	Raw timestamped process data	Clean, numeric, aligned feature matrix
Feature-engineering interface	Aligned process variables	Lagged/rolling/time features
Prediction interface	Feature vector at time t	Predicted % Silica Concentrate
Evaluation interface	Predictions + true lab values	MAE, RMSE, R^2 , residuals
Monitoring interface	Prediction + thresholds	Trend, risk flag, operator information

A practical implementation can be developed in Python using Pandas and NumPy for data preparation, Scikit-learn for regression pipelines and evaluation, XGBoost for boosted-tree modelling, Matplotlib for plots, and TensorFlow/Keras if an LSTM sequence model is added.

6 Performance Test

Performance testing is designed around industrial constraints rather than only academic model fit. The most important constraints are target delay, time alignment, leakage prevention, prediction horizon, computational cost, robustness to missing data and interpretability.

6.1 Test Plan / Test Cases

Test Case	Purpose	Metric / Check
TC-01 Data integrity	Verify schema, types and timestamp continuity	Rows, columns, null count, duplicates
TC-02 Baseline	Establish a simple reference	MAE, RMSE, R^2
TC-03 Model comparison	Compare Decision Tree / RF / Boosting	MAE, RMSE, R^2 , training time
TC-04 Minute-level prediction	Assess 1-minute target resolution	MAE/RMSE/ R^2 over chronological holdout
TC-05 Horizon study	Assess 1h, 2h, 3h+ forecasting	Error vs horizon
TC-06 No-Iron ablation	Remove % Iron Concentrate	Performance delta
TC-07 Leakage audit	Ensure future values are not used	Feature timestamp audit
TC-08 Robustness	Simulate missing/outlier sensor values	Performance degradation

6.2 Test Procedure

1. Sort all records by timestamp and preserve chronological order.
2. Define the prediction timestamp and the exact information cut-off available to the model.
3. Construct features only from observations at or before the cut-off.
4. Create chronological train, validation and test periods.
5. Fit preprocessing and model parameters only on the training data.
6. Tune hyperparameters using the validation period or time-series cross-validation.
7. Evaluate the final model once on the untouched test period.
8. Repeat the evaluation for different prediction horizons and feature configurations.
9. Record all metrics, model parameters and preprocessing assumptions.

6.3 Performance Outcome and Research Benchmarks

The internship report distinguishes between independently generated experimental results and published/reference results. Because the supplied materials did not include a runnable copy of the full 737,453-row CSV, this report does not invent a test-set score. Instead, published work is used to justify the model-selection strategy and expected behaviour.

Reference	Reported Insight	Relevance to This Internship
Moraes et al. (2018)	Soft-sensor study compared Random Forest, Gradient Boosted Trees and MLP; temporal lag was identified as an important challenge.	Supports lag engineering, model comparison and industrial soft-sensor framing.
Pu, Szmigiel & Apel (2020)	Stacked LSTM was used for hourly flotation data and reported stronger modelling capability than a Random Forest baseline.	Supports sequence modelling as a future/advanced approach.
Public project benchmark	One public implementation reported Random Forest as stronger on the large 20-second dataset and Gradient Boosting on hourly resampled data.	Supports testing both high-frequency and hourly representations; external benchmark only.

Recommended final performance table for the student's executed notebook: Model | Features | Horizon | MAE | RMSE | R^2 | Training Time. These values should be filled from the actual experiment before final academic submission if measured scores are required.

7 My Learnings

The internship strengthened both technical and professional skills. The most important technical lesson was that industrial machine learning is not limited to choosing an algorithm. Data understanding, temporal alignment, feature availability, validation strategy and deployment constraints can influence the result as much as the model itself.

- Python and Pandas for industrial-scale data loading, cleaning and transformation.
- Datetime handling, resampling and time-series feature engineering.
- Exploratory data analysis using distributions, trends, correlation analysis and residual diagnostics.
- Regression algorithms and model-selection principles.
- Hyperparameter tuning and the importance of avoiding overfitting.
- MAE, RMSE and R^2 as complementary evaluation measures.
- Feature importance and interpretation of model behaviour.
- Understanding data leakage in delayed-target industrial problems.
- Research reading and converting published findings into an implementation plan.
- Technical documentation, presentation and communication of an AI solution to an engineering audience.

From a career perspective, the project provided a practical example of how AI/ML can be applied to industrial data. It also demonstrated the importance of combining machine-learning knowledge with domain understanding, data engineering and software/deployment skills.

8 Future Work Scope

- Train and validate the full minute-level soft sensor using the complete CSV and a leakage-safe temporal split.
- Perform systematic forecast-horizon experiments from 1 hour onward and identify the point at which model error becomes operationally unacceptable.
- Implement LSTM/GRU/Temporal Convolution or Transformer-based sequence models and compare them with XGBoost.
- Add explainability using SHAP or permutation importance to support operator trust.
- Develop concept-drift monitoring so the model can detect changes in plant operating conditions.
- Create a Streamlit or web dashboard for live prediction and historical trends.
- Integrate the prediction service with an industrial IoT platform through a suitable API or messaging protocol.
- Add uncertainty estimation or prediction intervals so operators can distinguish high-confidence from low-confidence predictions.
- Test model robustness under sensor dropout, missing values and abnormal operating conditions.
- Retrain or recalibrate the model periodically using new laboratory measurements.

The most important future improvement is to move from offline prediction to a validated online soft sensor. Any production deployment should include strict timestamp validation, fail-safe behaviour, model monitoring and human review before automated control actions are allowed.

9 Conclusion

The six-week internship project studied quality prediction in an industrial iron ore flotation process using machine learning. The project reframed a delayed laboratory-quality problem as a time-series soft-sensor problem, with % Silica Concentrate as the prediction target.

The dataset is large, multivariate and time-dependent, containing approximately 737,453 observations and 24 columns over March–September 2017. Its mixed sampling rates make synchronization and feature availability central to the solution. The proposed pipeline therefore emphasizes data cleaning, temporal alignment, lagged features, chronological validation, model comparison and leakage audits.

Research indicates that classical ensemble models provide strong tabular baselines while sequence models such as LSTM can be valuable when temporal dependencies are explicitly represented. The internship provides a structured foundation for answering the original project questions through reproducible experiments, particularly the feasibility of minute-level prediction, the useful forecast horizon and the effect of removing % Iron Concentrate.

Overall, the project demonstrates how industrial AI can support earlier quality information, potentially improving process responsiveness, reducing waste and enabling more data-driven operation. The next stage is to execute the complete pipeline on the full dataset and validate the resulting soft sensor under realistic time-based test conditions.

10 References

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